

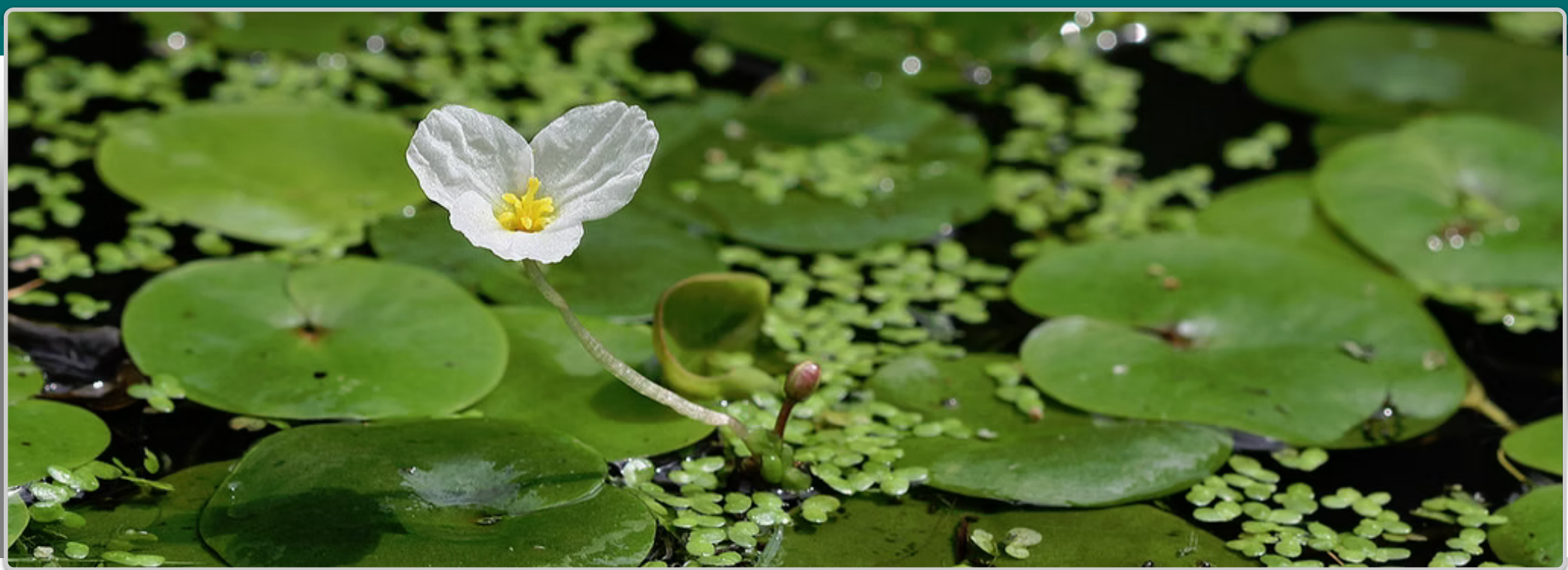


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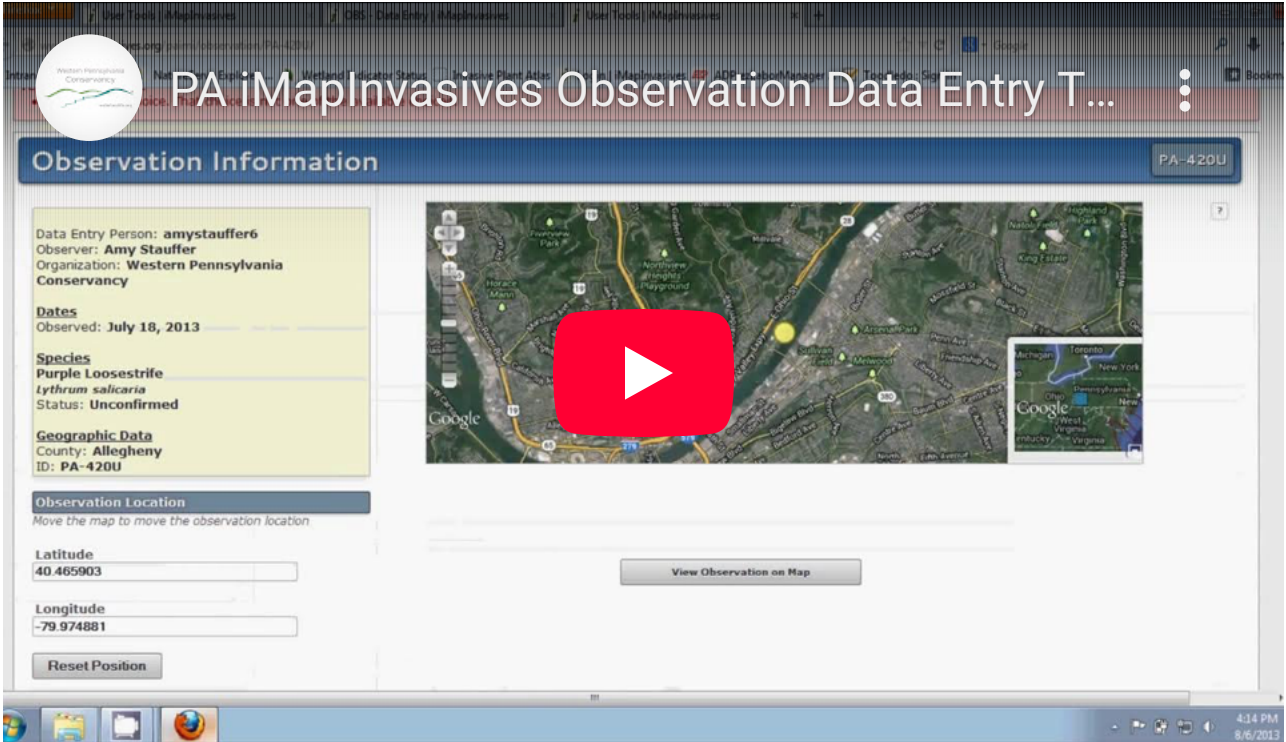
European Frog-Bit

(Hydrocharis morsus-ranae)

Report this Species!

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Species at a Glance

European frog-bit is an herbaceous aquatic plant that resembles a miniature water lily. It has been found in the Great Lakes since the 1930s but is now spreading inland into streams and lakes within the basin.

Identification

Leaves: Small, thick, heart-shaped, and leathery leaves are 1.5-6.5 cm (0.6-2.6 in) long and occur in clumps that are not anchored to the bottom sediment. They have smooth edges resembling those of a miniature water lily. A dark purplish-red spongy coating is present on the underside of the leaves, allowing it to float on the water's surface.

Flowers: Small, showy flowers are about 1 cm (0.4 in) in length and show up in early summer. Each flower has three white petals and a yellow center.

Stems/Roots: Stem-like extensions called stolons run from the center of the plant to produce juvenile plants. These stolons also produce turions that break free and sink to the bottom to lie dormant for the winter. Numerous free-floating, unbranched roots grow up to 30.5 cm (12 in) and form thick mats of tangled roots and runners.

Similar Species

While it is often mistaken for species of water lily, European frog-bit leaves are distinctly heart-shaped, leathery, and usually smaller. Water lily flowers are much larger with more than three petals. It may also be confused with American frog-bit, little floating heart, spatterdock, and watershield.

Habitat

European frog-bit prefers quiet, still, calcium-rich areas such as marshes, fens, swamps, backwaters, bays, sheltered coves, slow-moving shorelines of rivers, streams, and lakes, and poorly drained ditches.

Spread

European frog-bit can spread to new areas by plant fragments or by turions, which float to the surface and begin to grow in the spring. A single plant can produce 100 to 150 turions in one season. It can also hitchhike on boats, trailers, waterfowl, and flowing currents, and may also be **spread deliberately by humans** that purchase it as an aquarium or water garden plant.

Distribution

Native to Europe and northern Asia, European frog-bit was introduced intentionally in the United States as a commercial ornamental species. It escaped cultivation and spread to the Canadian shorelines of Lake Erie, Lake Ontario, the St. Lawrence River in New York, and Lake Champlain in Vermont. Populations are also present in Pennsylvania, New Jersey, Michigan, and Washington.

Note: Distribution data for this species may have changed since the publication of the [Mid-Atlantic Field Guide to Aquatic Invasive Species \(2016\)](#), the source of information for this description.

Environmental Impacts

European frog-bit populations increase rapidly, forming dense mats that decrease the amount of nutrients, dissolved oxygen, and light penetration into the water and limiting the growth of any native vegetation beneath. These mats can also inhibit the movement of waterfowl and fish, and limit recreational activities; however, it can serve as a food source for some species of water birds, fish, and insects.

Video



Note

Information for this species profile comes from the [Mid-Atlantic Field Guide to Aquatic Invasive Species \(2016\)](#).

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